Imperial College London
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Causal Structure Discovery via a Self-Implemented FCI$^+$ Algorithm:

A Case Study on the Tennessee STAR Project

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IRP repository:
https://github.com/ese-ada-lovelace-2025/irp-yq225

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Abstract

This project develops an auditable, independent implementation of FCI and FCI$^+$, validates both algorithms, and applies them to the Tennessee STAR class-size experiment. FCI learns Partial Ancestral Graphs (PAGs) under latent confounding and selection, but its Possible-D-SEP search is expensive; FCI$^+$ has a polynomial conditional-independence (CI) query bound for sparse graphs. The resulting fully typed Python package contains both paper-aligned schedules and structured decision traces, with no causal-discovery dependency. Validation combines 367 automated tests, three exact m-separation targets, a 15-run known-truth benchmark, and differential comparisons with causal-learn and R `pcalg`. Robust FCI$^+$ attains mean skeleton F1 0.985 and endpoint accuracy 0.840, versus 0.977 and 0.737 for paper-profile FCI. In 6,325 STAR kindergarten-cohort students, small classes show a 13.90-point observed kindergarten advantage (school-cluster 95% interval 5.21–22.07), whereas aide contrasts are near zero. This is not an identified treatment effect because analysed outcomes are post-assignment selected. Across methods, skeleton relations are more defensible than several order-, threshold-, or chronology-sensitive endpoints; no learned PAG identifies a unique DAG or effect magnitude.

1 Introduction

Problem description. Many empirical studies ask causal questions where not every common cause has been measured: family background, institutional selection, and administrative recording generate dependence that no causally sufficient DAG over the recorded columns represents honestly, and fitting one anyway turns uncertainty into unjustified arrows. FCI was designed for this setting: under causal Markov, faithfulness, and appropriate sampling assumptions, it learns from conditional-independence (CI) information a Partial Ancestral Graph (PAG) that remains valid under latent confounding and selection (Spirtes et al., 2000; Richardson and Spirtes, 2002; Glymour et al., 2019), at the price of an expensive Possible-D-SEP search and endpoint ambiguity; Claassen et al. (2013) prove a polynomial worst-case CI-query bound for sparse graphs of fixed maximum degree k.

The research problem is twofold: can FCI and FCI$^+$ be reimplemented faithfully and verifiably from their primary sources, and what can such an implementation legitimately conclude from a real randomized educational study — the Tennessee STAR class-size experiment — whose observed data are shaped by attrition and selection?

Significance. Public FCI-family tools typically return a finished graph with little visibility into why each endpoint was oriented; among the interfaces checked in section 2, none exposes a Python Claassen FCI$^+$ with paper-profile control and a structured audit trail. An implementation whose every mark traces to recorded CI decisions serves applied researchers defending structural claims, methodologists needing a typed reference for differential testing, and instructors separating design-based from discovery evidence. Substantively, STAR remains a canonical evidence source on class size, and selective attrition in longitudinal experiments is a live methodological concern.

Objectives and contributions. Four measurable objectives respond directly to the auditability and finite-sample gaps above; each also defines a project contribution.

O1/C1. *Implement source-faithful algorithms.* Success requires separate paper profiles whose stages map to primary-source concepts, explicit PAG endpoint semantics, typed public APIs, and no causal-discovery dependency. The delivered 9,559-line, 38-module package supplies five CI tests, ordered Possible-D-SEP, complete Algorithm 2 stages, and five orientation schedules.

O2/C2. *Verify and validate.* Success requires exact recovery of all pre-specified oracle targets, finite-sample accuracy reporting, and a clean automated test matrix. The apparatus comprises an exact MAG m-separation oracle, three published-structure targets, 15 known-truth runs, and 367 tests under strict typing on Python 3.9–3.13.

O3/C3. *Position against established software.* Success requires matched-input comparisons with disclosed interface mismatches, using skeleton, endpoint, runtime, and CI-query metrics. Executed causal-learn and R `pcalg` runners provide this evidence.

O4/C4. *Deliver a defensible STAR application.* Success requires independent R replication plus school-cluster bootstrap, cyclic-order, chronology, and sensitivity audits, with conclusions promoted only through a pre-specified evidence hierarchy.

C1 and C4 are deliberately separated: the algorithm stays reusable and domain-neutral, while the application states its own coding, selection, clustering, and interpretation decisions.

These objectives answer one overarching question: can source-faithful FCI and FCI$^+$ be implemented and validated well enough to support bounded, transparent structural claims on STAR? Section 2 establishes the source and software gap; sections 3 and 4 test O1–O4; sections 5 and 6 distinguish supported conclusions from exploratory patterns. Controlled evidence precedes the application, so a real-data pattern never serves as evidence of algorithm fidelity.

2 Preliminaries and Related Work

2.1 Graphs, assumptions, and PAGs

Constraint-based discovery begins from statements $X \perp Y \mid Z$ over observed variables V, with latent variables L and selection variables S possibly present. The *causal Markov condition* states that graphical separation implies conditional independence; *faithfulness* states that observed independences arise from separation rather than exact cancellation. Separation in a DAG is d-separation:

Definition 1 (d-separation)**.** A path p is blocked by a node set Z if and only if p contains a chain or fork whose middle node is in Z, or a collider whose middle node is neither in Z nor an ancestor of Z. If Z blocks every path between X and Y, they are d-separated given Z (Pearl, 2009).

Under causal sufficiency these assumptions support DAG methods such as PC. Otherwise the appropriate observed-variable object is a maximal ancestral graph (MAG) (Richardson and Spirtes, 2002): marginalising L and conditioning on S induce directed, bidirected, and undirected relations, and every missing MAG edge corresponds to an *m-separation*, the generalisation of d-separation to mixed ancestral graphs and the criterion used by this project's exact oracle. Because different MAGs can encode the same independence model, the identifiable output is a PAG, recording exactly the endpoint features shared by one Markov equivalence class (Zhang, 2008).

Each PAG adjacency carries two endpoint marks: a circle leaves ancestry unresolved, an arrowhead at Y excludes Y as an ancestor of X, a tail asserts ancestry in every represented MAG, a bidirected edge is compatible with latent confounding without identifying a unique latent variable, and undirected edges reflect selection structure. An adjacency says only that the tested sets failed to separate the pair; neither an adjacency nor an endpoint alone identifies a structural coefficient, an adjustment set, or an interventional contrast (Pearl, 2009). Sample CI tests are error-prone, so exact-oracle recovery isolates graph logic rather than finite-sample accuracy. Among the discovery families reviewed by Glymour et al. (2019), score-based search (such as GES) and standard functional causal models (such as LiNGAM) assume causal sufficiency, so the constraint-based FCI family is the appropriate class when latent confounding and selection must be represented.

2.2 Standard FCI and its historical boundary

FCI does not require that all common causes of measured variables are measured (Spirtes et al., 2000, Ch. 6). A PC-style search removes $X - Y$ when some subset of current neighbours separates the pair, but with latent variables a separable pair may stay adjacent: a necessary separator can contain a node no longer adjacent to either endpoint. The book introduces D-SEP to characterise where an observed separator can reside and, since D-SEP is not computable from a partial graph, an observable superset:

Definition 2 (Possible-D-SEP)**.** For an ordered pair (A, B) in a partially oriented graph, a node $V \neq A$ belongs to Possible-D-SEP(A, B) if some path from A to V has every internal triple either a collider or a triangle whose middle node is not a definite noncollider (Spirtes et al., 2000).

The construction is deliberately permissive and required *separately* from each ordered endpoint: using one direction only, or mixing pools, changes the search contract. The full schedule: PC-style adjacency search;

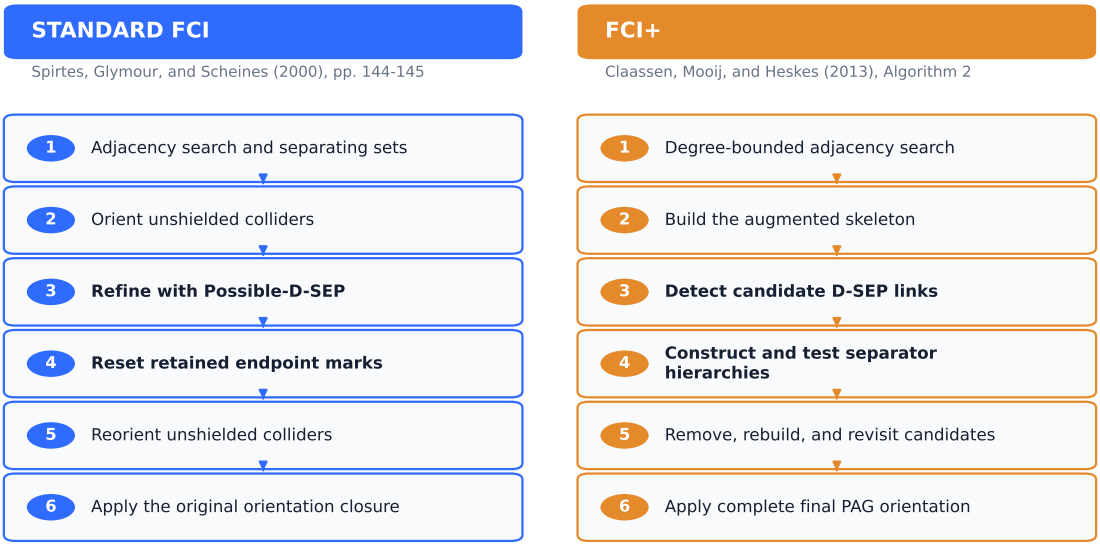


Figure 1: Source-located control-flow comparison. Standard FCI performs a non-local Possible-D-SEP refinement; FCI+ instead iterates augmented-skeleton, candidate D-sep-link, hierarchy, and minimal-separator stages before invoking final orientation. The robust application profile of section 3 changes finite-sample policy, not the definition of Algorithm 2.

initial unshielded colliders; ordered Possible-D-SEP refinement; endpoint reset and collider reconstruction from final separators; orientation closure. Theorem 6.4 of the source guarantees a correct *partially oriented inducing path graph* under correct population-level CI decisions — while the authors state they did not know whether the output was maximally informative. Complete orientation therefore depends on the later rules of Zhang (2008) and is not retroactively attributed to the 2000 source.

2.3 FCI+: Algorithm 2

Claassen et al. (2013) ask whether causal discovery remains intrinsically exponential when the observed structure is sparse. Their central result, restated in the form used throughout this report:

Theorem 1 (Claassen et al., 2013)**.** Assume a constant-time exact independence oracle, a distribution faithful to a causal DAG, and an observed-variable MAG of maximum degree at most a constant k. Then Algorithm 2 (FCI+) returns the completed PAG over N observed variables using $O(N^{2(k+2)})$ CI queries in the worst case.

The claim is narrower than the paper's title: it bounds oracle queries and does not make a statistical test exact or constant-time. A nonadjacent pair in the true MAG can survive bounded adjacency search when every useful separator requires a node outside the endpoint-adjacency union; such a false adjacency is a *D-sep link*. Rather than searching a large Possible-D-SEP pool, FCI+ extracts invariant non-ancestry into an *augmented skeleton* G^+: when a known minimal independence $X \perp Y \mid S$ becomes dependence given $S \cup \{A\}$ for a suitable adjacent A, an invariant arrowhead at A is supported. In G^+, a D-sep link $X - Y$ appears as the middle edge of $U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$ with U, V nonadjacent and cross-direction paths not running against an arrowhead; the pattern only *proposes* a candidate, and independence evidence decides deletion. Conditioning sets come from the recursive hierarchy $\mathrm{HIE}(X, I)$, the fixed point of repeatedly adding recorded minimal separators of members; because candidates can depend on one another, each successful removal rebuilds the augmented skeleton and candidate list (figure 1). The arXiv supplement's abstract states $N^{2(k+1)}$ while its theorem and proof agree with the UAI paper's $N^{2(k+2)}$, a discrepancy recorded rather than harmonised. Algorithm 2 invokes complete FCI orientation rather than rederiving it, so its completed-output claim inherits Zhang (2008) as a dependency.

Table 1: Dated public-interface capability comparison (26 July 2026). All four tools expose standard FCI; "partial" means some information is obtainable, but not through the same integrated result contract. Tetrad is documentation-only because no working Java runtime was available.

Tool	Claassen FCI+	Custom CI	Sepset provenance	Order/bootstrap audits	Structured audit and export
`fci_engine` 0.1.0	yes	yes	yes, stage named	both	CI and orientation traces; D-SEP diagnostics; per-edge explanations; JSON/CSV/HTML
`pcalg` 2.7-12	yes	yes	partial	order partial	standard-FCI sepsets; no equivalent bundled export established
causal-learn 0.1.4.7	no public entry point found	named tests	not exported	not established	graph return; no equivalent structured trace established
Tetrad 7.6.10	not established	yes	yes	partial, broader workflow	logger and GUI facilities; not executed here

2.4 What the sources do not establish

Neither source guarantees finite-sample recovery: both theorems are conditional on correct CI decisions. Neither identifies a unique DAG, an adjustment set, or a treatment effect; a bidirected pattern does not name or measure a hidden cause. The 2000 source proves no maximal orientation and no polynomial bound; the 2013 source proves no finite-sample superiority and nothing when k is wrong — a user-entered bound is not evidence — and its authors warn that hierarchy-derived conditioning sets may be large, reducing finite-sample power.

2.5 Existing software

Table 1 summarises the dated capability comparison: `pcalg` (Kalisch et al., 2012) exposes both algorithms and accepts user-defined CI tests, but its `fciPlus` result carries materially less separator provenance; causal-learn (causal-learn developers, 2026) exposes standard FCI but no public Claassen FCI+ entry point; Tetrad (Tetrad Project, 2026) provides FCI-family hybrids, none established as a drop-in Claassen FCI+, and was checked at documentation level only (no working Java runtime). The originality of this work is therefore not a new algorithm but a verifiable one: paper-profile control, structured traces, separator provenance, per-edge explanations, and stability auditing integrated in one workflow — an auditability improvement, not a universal accuracy or speed claim.

3 Methodology

3.1 Package design

The package exposes the obligations of section 2 as inspectable components, separate from application policy: reusable code in `src/fci_engine`, the STAR workflow in `case_studies/tennessee_star`, no STAR-specific choice in the estimator. The algorithms are separate entry points with explicit paper profiles, so a finite-sample default cannot silently stand in for the source algorithm. `FCIConfig.paper()` fixes unbounded depth, immediate deletion, first-found separators, enabled Possible-D-SEP, nonconservative colliders, and the historical `spirtes_2000` closure; `FCIPlusConfig.paper(k)` requires an explicit k for both conditioning depth and sparsity bound, disables Possible-D-SEP, and selects the complete `standard` closure required by Algorithm 2's output claim. Profiles constrain execution; they do not validate theorem assumptions — supplying $k = 3$ does not prove the true MAG has degree at most three.

The `PAG` class stores an endpoint mark at each side of every adjacency, keeping tails, arrowheads, circles, and absence explicit. Every search uses one `CITest.test(data, x, y, cond_set)` contract, with built-in Fisher-Z, missing-value Fisher-Z, chi-square, G-square, and kernel tests. A cache canonicalises each query, separating the *logical* CI requests made by control flow from cache hits and executed statistics — the distinction relevant to a theorem counting oracle queries.

`FCI.fit` exposes the source stage ordering, recording Possible-D-SEP removals with source `pdsep`. `FCIPlus.fit` deliberately does not call Possible-D-SEP; it passes the initial skeleton to the Algorithm 2

search: augmented skeleton, bidirected witness patterns, paired endpoint bases, hierarchy fixed points, separator testing and minimisation, graph update, and candidate rebuild. A diagnostics record counts candidates, revisits, hierarchy queries, D-SEP-stage CI requests, and the maximum conditioning size — empirical accounting, not a mechanised proof of the bound. One dispatcher provides five orientation schedules: `spirtes_2000`, the historical closure used by the standard-FCI paper profile; `standard`, the modern complete schedule required by the FCI+ paper profile; and `conservative`, `leaf`, and `robust`, finite-sample engineering policies supporting no completed-PAG claim. A regression verifies that `spirtes_2000` stops before a selection-bias tail completion that `standard` orients, making the historical boundary executable. `FCIResult` binds the final graph to the evidence that produced it — configuration, separators with stage sources, logical CI counts, and named CI and orientation traces; `explain_edge` returns every decision behind a pair, so a plotted arrow cannot detach from the profile, test, threshold, and rule that created it.

The code base was written entirely for this project: a 9,559-line, 38-module algorithm package; 367 tests in 44 modules (5,929 lines); a 3,052-line STAR application with a 396-line independent R runner; and benchmark and figure scripts — roughly 25,300 lines in total. The package ships `py.typed` and passes Ruff, strict MyPy, a wheel build, and Pytest on Python 3.9–3.13; a full concept-to-symbol-to-test traceability matrix is maintained in the repository's extended report. These checks establish interface and packaging consistency, not faithfulness, true sparsity, or any empirical causal conclusion.

The practical profile trades computation for stability: stable deletion reduces order dependence; max-p separators spend more CI queries for stronger independence evidence; conservative collider and tail policies preserve circles; automatic α is a documented sample-size heuristic. Accordingly, "paper FCI+" is the fidelity and query-efficiency subject; "robust FCI+" is the preferred application policy, meeting the stated audit criteria while returning less-specific endpoints — a local preference, not a universal ranking.

3.2 Validation and comparison design

The strongest tests replace sample CI decisions with exact m-separation in a known MAG, isolating search correctness from sampling error. Three independently specified targets come from published structures: the canonical Figure 4(b) D-SEP example of Claassen et al. (2013), a latent-variable reference graph from the FCI literature, and an orientation-rule reference graph associated with Zhang (2008). Finite-sample behaviour is then tested on 15 deterministic latent-SEM cases — five graph families, three seeds each, 2,500 observations, known oracle PAGs.

The software comparison separates execution from interface inspection: the capability layer appears in section 2; the executed layer asks what happened on identical committed data under recorded settings. The benchmark reuses the 15 known-truth runs, every method receiving the same numeric matrix and variable order. Fixed paper and reference rows use Fisher-Z at $\alpha = 0.001$; robust FCI+ uses its documented sample-size rule, which resolves to $\alpha = 0.01$ here — intentional: that row evaluates the public robust policy, not a literal Algorithm 2 reproduction. Exact F1 requires both endpoint marks to match; semantic F1 additionally credits endpoint patterns compatible with the target ancestral constraints; endpoint accuracy scores individual marks. The common conversion represents each edge as an unordered pair plus two semantic marks, never as generic directed edges. Throughout, data, order, alpha, and CI semantics are held fixed where interfaces allow, every mismatch is disclosed, and distinct FCI-family algorithms are never relabelled as one.

3.3 STAR data and panels

The Tennessee Student/Teacher Achievement Ratio project was a large randomized experiment in kindergarten through grade 3, assigning students and teachers to small classes (about 13–17 students), regular classes (about 22–25), or regular classes with a full-time aide (Word et al., 1990); it remains a central source of evidence on class size (Krueger, 1999). This analysis uses the official Harvard Dataverse release (Achilles et al., 2008): 11,601 student records, 379 variables, CC0 licence, codes checked against the 146-page user guide.

Randomization already provides a stronger basis for estimating the class-size effect than structure learning, so FCI is not used to rediscover it. The purposes are structural: the attrition structure of grade-3 observation;

Table 2: STAR analysis panels.

Panel	Rows	Nodes	Purpose
Attrition / observation	5,744	9	Relate assignment, characteristics, context, and kindergarten achievement to whether both grade-3 scores are observed.
Longitudinal achievement	2,787	9	Study structure between kindergarten and grade-3 achievement among complete cases while retaining early achievement.
Focused treatment-outcome	2,976	8	Study the class-assignment/grade-3 relationship without allowing kindergarten achievement, an early post-assignment outcome, to serve as a separator.

the persistence of the early–later achievement adjacency; the effect of an early post-assignment outcome's inclusion; cross-implementation agreement; and FCI$^+$'s search savings. The design also supplies an external audit: a data-only PAG implying an implausible pre-treatment direction, or latent confounding of a randomized assignment, should be reported as a discrepancy rather than accepted literally.

The analysis covers the 6,325 students with a kindergarten class assignment across 79 schools, arranged into three panels deliberately not pooled because each encodes a different statistical question (table 2). All model inputs are integer-coded discrete categories: class arm, sex, ethnicity (small source categories collapsed), free-lunch status, school context, entry-age quantiles, binned teacher experience, and reading-plus-mathematics scaled scores converted to achievement quartiles; the exact numeric tables supplied to the algorithms are committed in the repository.

3.4 Discovery protocol and audits

Before discovery, randomized-arm contrasts are computed for kindergarten scores, grade-3 scores among students with both scores, and the probability that both are observed, with uncertainty from 1,000 kindergarten-school bootstrap samples; these design-based contrasts are reported separately from discovery output.

Discovery uses the likelihood-ratio G-square test, $G^2 = 2\sum_s \sum_{x,y} O_{xys} \log(O_{xys}/E_{xys})$, at $\alpha = 0.05$ — chosen because every panel variable is an integer-coded category, where Fisher-Z's Gaussian-linear assumptions do not apply, at the cost of sparse contingency cells under high-order conditioning. Standard FCI runs its unbounded paper schedule; paper FCI$^+$ uses $k = 3$; robust FCI$^+$ uses the practical profile (stable skeleton, strongest separating sets, conservative colliders, robust orientation, depth and sparsity bound 3); the independent reference is `pcalg::fciPlus` with selection bias enabled and no exposed k. No temporal priors are supplied to discovery; reversals are audited after fitting.

Alternatives were rejected for stated reasons: PC and GES impose causal sufficiency; Fisher-Z mismatches the discrete panels; kernel CI is far more expensive and does not solve clustered sampling; and adding temporal priors would conceal, rather than test, chronology failures. A formal randomized treatment-effect model is complementary to the structural question and is therefore kept outside the discovery analysis.

Three audits probe stability: 100 school-cluster bootstrap refits per algorithm and panel, recording each unordered adjacency's appearance frequency (an exploratory diagnostic, not a posterior probability); a cyclic-order audit refitting every cyclic shift of the variable list for all profiles and the R reference; and a focused-panel sensitivity grid over $\alpha \in \{0.01, 0.05\}$, three versus four quantile bins, and $k \in \{2, 3, 4\}$.

A skeleton-level conclusion is promoted only when the adjacency survives every cyclic shift with high bootstrap frequency; a directional conclusion additionally requires invariant endpoint marks, no conflict with measurement time or randomization, and no material reversal under sensitivity checks; failure downgrades the claim to unresolved direction rather than deleting the edge.

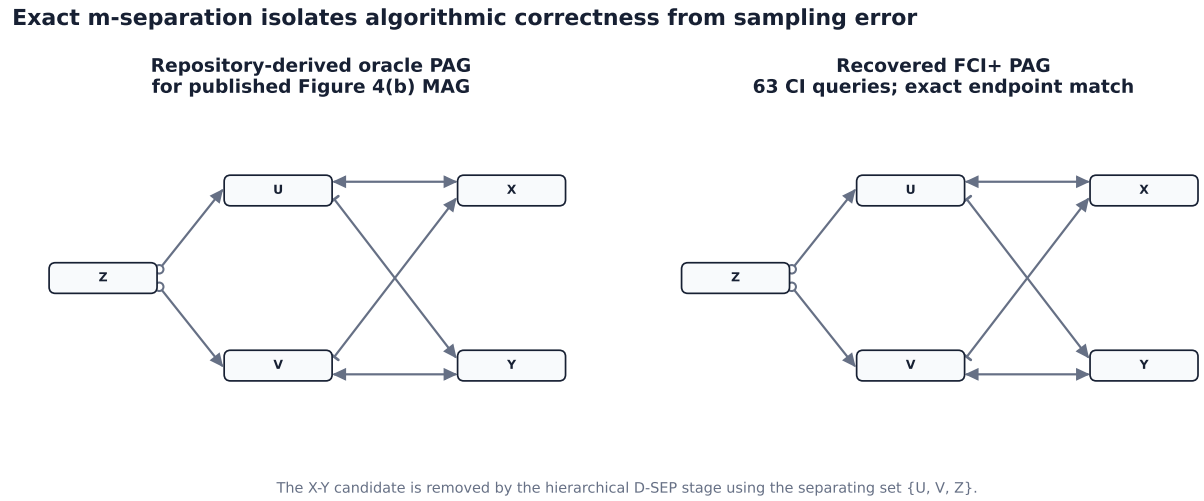


Figure 2: Exact-oracle validation on the central Figure 4(b) example. The source-grounded facts are the six MAG edges and the exhibited separator $\{U, V, Z\}$; the endpoint-complete comparison target is independently derived in the repository. Paper-aligned FCI needs 102 oracle queries, FCI+ 63; both recover the target exactly.

Table 3: Executed known-truth comparison: means over 15 completed runs (five latent-SEM families, three seeds each, 2,500 observations).

Method (version)	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	Seconds	Logical CI
Local FCI paper (0.1.0)	0.977	0.532	0.943	0.737	0.020	449.7
Local FCI+ paper (0.1.0)	0.979	0.536	0.951	0.744	0.009	197.1
Local FCI+ robust (0.1.0)	0.985	0.730	0.985	0.840	0.018	401.9
causal-learn FCI (0.1.4.7)	0.979	0.536	0.951	0.744	0.009	not exposed
R `pcalg` FCI+ 2.7-12	0.979	0.536	0.951	0.744	0.500	not exposed

4 Results

4.1 Exact-oracle validation

Both algorithms recover all three encoded targets, with Figure 4(b) additionally tested under four variable orders. The Figure 4(b) MAG has six edges ($Z \to U$, $Z \to V$, $U \leftrightarrow X$, $V \to X$, $U \to Y$, $V \leftrightarrow Y$); the adjacency stage retains a false $X - Y$ link although $X \perp Y \mid \{U, V, Z\}$, with Z adjacent to neither endpoint. The source exhibits this separator but prints no endpoint-complete PAG, so the completed comparison target is repository derived (figure 2). Here paper-aligned standard FCI needs 102 CI queries and FCI+ 63 (a 38.2% reduction), both returning the exact target; the values are frozen in `reports/data/oracle_validation_summary.csv` with target hash and separator provenance.

4.2 Finite-sample known-truth benchmark

On a seeded latent linear SEM over the same reference graph, at $N = 5{,}000$ FCI+ returns an extra $X \leftrightarrow Y$ (exact-edge F1 0.923) while standard FCI recovers the exact PAG; at $N = 50{,}000$ both are exact — a reproducible fixture, not a universal threshold.

Across the 15 cases (table 3, first three rows), the robust profile improves the mean skeleton and endpoint scores at about 2.04 times the CI cost of paper FCI+ — not proof of universal dominance, but a check that the STAR stability gain reported below is not bought by discarding structure, and a warning that fewer CI tests do not imply better accuracy: the algorithms take different logical routes after the initial skeleton and make different errors. Oracle tests support pipeline correctness under exact CI information; finite-sample results remain limited by CI-test error and tuning.

Table 4: Observed outcome summaries by kindergarten class arm. Intervals use 1,000 kindergarten-school bootstrap samples. Grade-3 summaries condition on both scores being observed; kindergarten summaries also exclude missing score components, so both are design-supported observed-outcome comparisons rather than a fully reconstructed intention-to-treat estimand.

Outcome	Arm	n	Estimate	95% interval
Kindergarten score	Regular	2,005	918.04	908.78 to 927.05
	Regular + aide	2,043	918.36	908.52 to 927.07
	Small	1,738	931.94	922.70 to 940.12
Grade-3 score	Regular	1,047	1246.74	1238.96 to 1254.01
	Regular + aide	1,019	1247.25	1238.53 to 1255.48
	Small	936	1258.43	1250.58 to 1266.17
Grade-3 observed	Regular	2,194	47.72%	42.72% to 52.14%
	Regular + aide	2,231	45.67%	40.64% to 50.39%
	Small	1,900	49.26%	44.46% to 53.77%

4.3 Comparison with established software

The equality of the aggregate local paper-FCI⁺, causal-learn, and `pcalg` scores in table 3 is useful differential evidence, not an equivalence proof: local and causal-learn timings are in-process whereas the R timing includes process startup, and `pcalg::fciPlus` exposes no identical degree-bound control. The executed layer thus supports C3: matched aggregate behaviour where interfaces overlap, alongside the integration gap documented in section 2.

4.4 Observed randomized-arm outcomes

The small-minus-regular kindergarten contrast is 13.90 points (school-cluster 95% interval 5.21 to 22.07), while the regular-with-aide contrast is 0.31 points (−7.40 to 7.38). At grade 3 among observed students the small-class contrast remains 11.69 points (3.74 to 19.38) and the aide contrast 0.51 (−5.70 to 7.22). Observation rates show no precise arm difference: small minus regular is +1.54 percentage points (−1.73 to 4.56), aide minus regular −2.05 (−5.35 to 1.41) (table 4).

4.5 Structure discovery across the three panels

On the attrition panel the three implementations substantially agree at the skeleton level (figure 3; costs and agreement in table 5). All three omit a direct `K_Class`–`Grade3_Observed` adjacency, which appears in only 12–16% of school bootstraps. By contrast, `K_Achievement`–`Grade3_Observed` is adjacent in 100% of bootstrap fits for all three local profiles and the free-lunch/observation adjacency is also stable: later availability is not independent of early achievement and measured context. The three endpoint readings diverge: standard FCI gives the impossible *grade-3 observation → kindergarten achievement*; self-FCI⁺ leaves a circle; R-FCI⁺ gives the temporally admissible direction. The R reading is more plausible, but package identity cannot establish statistical correctness. The attrition PAG does not prove a missingness mechanism; a formal analysis would need an explicit estimand and further assumptions.

On the longitudinal panel all three implementations retain the kindergarten/grade-3 achievement adjacency, present in all 100 bootstrap samples for every local profile. No full-data PAG retains a direct class-assignment/grade-3 adjacency once kindergarten achievement is included, yet that adjacency reappears in 71% of FCI, 79% of paper FCI⁺, and 80% of robust-profile bootstrap fits — a warning that conservative orientation and order invariance do not remove school-resampling sensitivity. Both FCI⁺ implementations orient grade-3 achievement toward kindergarten achievement, which is temporally impossible if read causally, while standard FCI leaves that endpoint unresolved. Agreement between implementations is not sufficient to override chronology: both may replicate the same finite-sample CI pattern under complete-case selection.

Removing kindergarten achievement changes the result: all three algorithms return the same 10-edge PAG

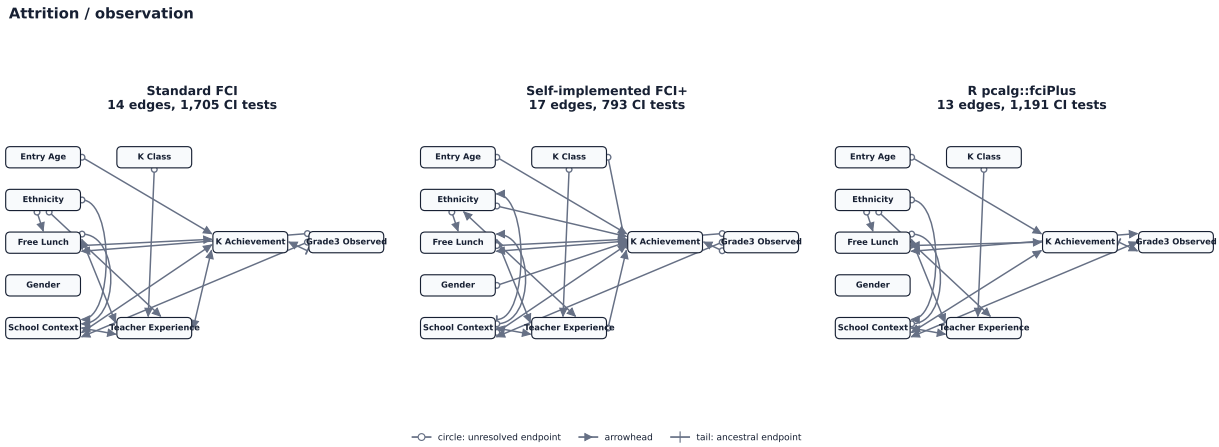


Figure 3: Three PAGs for the attrition/observation panel: self-implemented FCI, self-implemented FCI⁺, and R `pcalg::fciPlus`. The skeletons substantially overlap, while the achievement/observation endpoint differs across implementations.

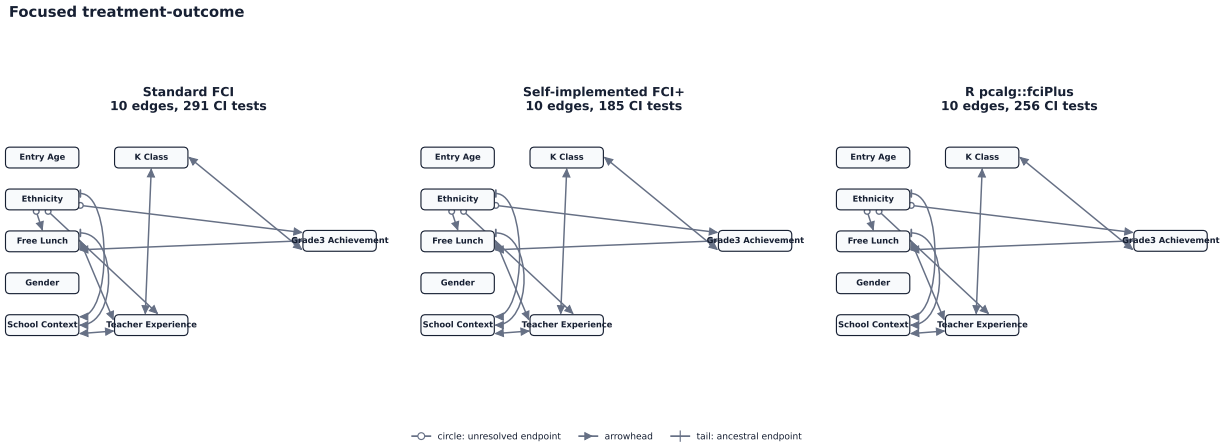


Figure 4: PAGs for the focused complete-case panel. All three algorithms recover the exact same 10-edge skeleton and endpoint pattern, including the bidirected class-assignment/grade-3 edge.

— not merely the same skeleton — including *K_Class* ↔ *Grade3_Achievement* (figure 4), an adjacency appearing in 92%, 85%, and 96% of FCI, paper, and robust bootstraps. Independent exact replication makes this conditional-independence pattern more credible, but it does not make a literal latent-confounding reading credible: kindergarten class assignment was randomized, while the focused panel conditions on grade-3 outcomes being observed and omits early post-assignment achievement. Selection on observation, school-level dependence, omitted post-randomization variables, and CI-test error can all produce arrowheads at both endpoints.

The sensitivity result is decisive (figure 5): the focused adjacency is robust to the tested bin count and degree bound but not to the CI threshold: at $\alpha = 0.01$ no local configuration retains it, while at $\alpha = 0.05$ the paper profiles retain a bidirected mark for every tested k and the robust endpoint keeps circles. The stricter threshold demands more evidence against independence, so removal indicates the edge rests on relatively weak conditional-dependence evidence rather than an invariant feature.

4.6 Robust application profile, agreement, and cost

The robust profile is a finite-sample decision layer, not a fourth paper implementation. It returns 17, 12, and 10 edges on the three panels, with target relations *K_Achievement* $\circ\!-\!\circ$ *Grade3_Observed*, *K_Achievement* $\circ\!-\!\circ$ *Grade3_Achievement*, and *K_Class* $\circ\!-\!\circ$ *Grade3_Achievement*. It retains the target skeleton information without the chronology reversal or the bidirected over-interpretation.

In the cyclic-order audit, paper self-FCI⁺ reproduces the exact full-data PAG in 9 of 9 attrition, 5 of 9

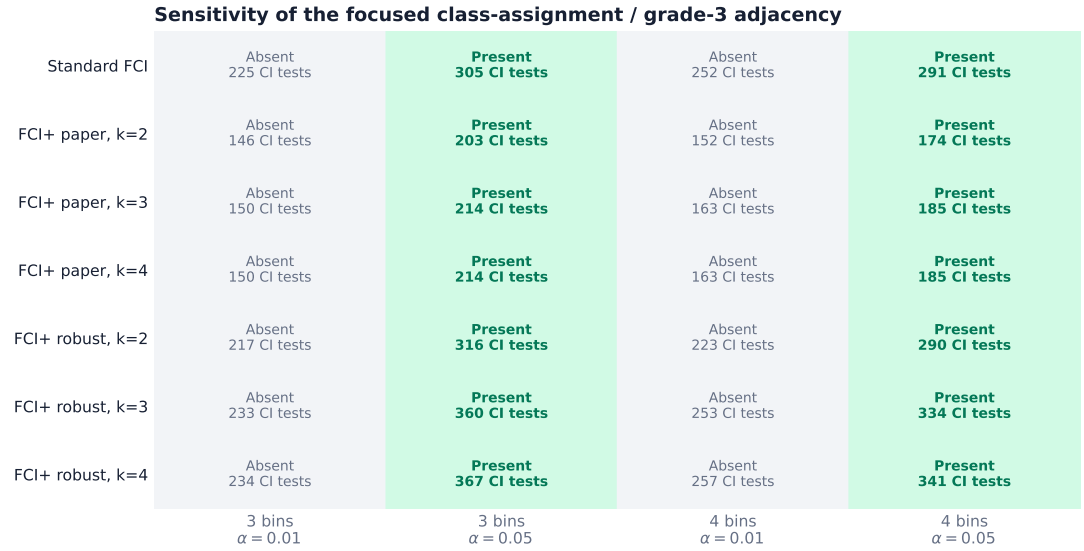


Figure 5: Focused treatment-outcome sensitivity. The adjacency appears at $\alpha = 0.05$ and disappears at $\alpha = 0.01$, regardless of whether three or four quantile bins or $k = 2, 3, 4$ is used for FCI⁺.

Table 5: Three-algorithm structural agreement (left) and search cost (right). Endpoint rates use only adjacencies shared by the indicated pair; time is the median of three full-data repetitions.

Panel	Pair	Jaccard	Endpoints
Attr.	FCI / self	0.824	50.0%
	FCI / R	0.929	61.5%
	self / R	0.765	38.5%
Long.	FCI / self	0.769	50.0%
	FCI / R	0.833	70.0%
	self / R	0.923	83.3%
Focus.	FCI / self	1.000	100.0%
	FCI / R	1.000	100.0%
	self / R	1.000	100.0%

Panel	Algorithm	Edges	CI	Sec.
Attr.	FCI	14	1,705	5.437
	self paper	17	793	1.788
	self robust	17	1,304	2.838
	R-FCI⁺	13	1,191	10.394
Long.	FCI	10	960	2.148
	self paper	13	335	0.337
	self robust	12	619	0.562
	R-FCI⁺	12	475	1.650
Focus.	FCI	10	291	0.322
	self paper	10	185	0.159
	self robust	10	334	0.247
	R-FCI⁺	10	256	0.884

longitudinal, and 6 of 8 focused shifts, with one longitudinal temporal reversal; the R reference preserves every panel skeleton while matching exact endpoints in 7 of 9, 6 of 9, and 6 of 8 shifts; robust self-FCI⁺ returns the identical PAG in all 26 refits with no reversal. Table 5 summarises agreement and search cost. Paper FCI⁺ uses 46.5%, 34.9%, and 63.6% of the standard-FCI CI-test counts on the three panels (R reference: 69.9%, 49.5%, 88.0%). Runtime is not language neutral, so CI-call counts are the more interpretable comparison. Relative to paper self-FCI⁺, the robust profile uses 1.64 to 1.85 times as many CI calls, purchasing stable deletion and invariant conservative endpoints rather than speed. In summary, the robust profile returns the same full PAG for every tested order with every target adjacency intact and no backward-in-time directed edge — at the price of under-orientation and extra CI work. Reliability has improved; identifiability has not.

5 Discussion

5.1 Evidence hierarchy

An explicit hierarchy prevents a visually definite PAG mark from outranking the experimental design. Tier 1: randomized assignment with the school-cluster contrast, permitting a bounded causal statement for the analysed observed outcome. Tier 2: an adjacency replicated across methods, resamples, orders,

and sensitivity settings — a robust skeleton-level claim with no magnitude or direction. Tier 3: a stable skeleton with circles or endpoint disagreement, where adjacency is more credible than orientation. Tier 4: an implementation-, order-, or chronology-sensitive endpoint — diagnostic evidence only. Tier 5: a threshold-sensitive or complete-case exploratory pattern — a hypothesis for external validation.

5.2 What can and cannot be concluded

The Tier-1 result — the 13.90-point small-class contrast — is consistent with a beneficial early effect, but it is not an identified causal effect in the post-assignment observed subset without assumptions about missing outcomes, compliance, and the original randomization blocks. The aide contrasts are close to zero at both time points: no evidence that an aide reproduces the small-class advantage. The achievement–achievement and achievement–observation adjacencies replicate across all fits and bootstraps: a robust *skeleton-level* relationship between early performance, later performance, and later availability (Tier 2). The full-data attrition PAGs retain no direct class-arm/observation adjacency and its bootstrap frequency is low: concern about selective attrition is supported without a stable direct arm effect on observation.

Conversely: no unique DAG is identified — every PAG represents an equivalence class — and no effect magnitude comes from FCI. A bidirected edge is not proof of a hidden confounder; in the focused panel it conflicts with the known randomization unless selection or post-treatment conditioning is considered. Temporal reversals are algorithm diagnostics, not credible chronology. Implementation agreement is not proof: shared CI decisions and rules can reproduce the same wrong endpoint. The 26 recorded shifts are a finite audit, not a proof of order invariance; bootstrap frequency is not a posterior probability. Absence is policy conditional: failure to retain an edge under one node set, test, and threshold is not proof of zero effect.

5.3 Limitations of the method

FCI-family guarantees require faithfulness and correct CI decisions; the threshold-sensitive focused edge demonstrates the practical consequence. The true STAR-MAG degree is unobserved, so $k = 3$ is an analysis setting, not a verified sparsity assumption. Discretisation enables G-square testing but discards within-bin information and creates sparse conditional tables. The row-wise CI test is not cluster-robust; school bootstrapping diagnoses sensitivity to school composition but does not repair that mismatch. Finally, version 0.1.0 and hand-authored oracle targets provide strong regression evidence, not a formal correctness proof or exhaustive MAG-equivalence check.

A deterministic scaling audit (5 to 80 nodes; a directed chain and the Figure 4(b) core with isolated variables added) recovers the exact target skeleton in every run but yields an unexpected finding — the query advantage is not universal: FCI$^+$ wins on the five-node core (62 versus 101 queries) yet spends 5,162 versus 3,251 at 80 nodes, and 24,338 versus 12,638 on the 80-node chain — consistent with a worst-case bound; augmentation has a real cost when targeted D-SEP search offers little benefit.

5.4 Limitations of the STAR study

The longitudinal and focused panels condition on observed grade-3 scores and may open selection paths. Kindergarten achievement is post-assignment: including it may block part of the class-size pathway, while excluding it may leave post-treatment dependence. Reporting both panels exposes this choice but does not identify either model as correct. Original randomization blocks, class switching, compliance, and school-specific procedures are not reconstructed; the composite score also leaves subject-specific and subgroup effects unexamined.

6 Conclusion and Future Work

All four objectives were achieved, with explicit scope qualifications. **O1 was met**: the package implements source-located FCI and Algorithm 2 profiles with typed, auditable results. **O2 was met**: all three exact targets were recovered, 367 tests pass, and the 15-run benchmark quantifies finite-sample error. **O3 was met with interface limits**: matched causal-learn and `pcalg` runs reproduce aggregate paper-profile scores, although neither exposes every control or diagnostic. **O4 was met within its structural scope**: the STAR workflow completed all planned replication and stability audits, but it identifies neither a formal intention-to-treat effect nor a unique causal graph.

The STAR evidence supports a 13.90-point observed kindergarten small-class advantage, no comparable aide advantage, stable early–later achievement structure, and selective later observation. The independently replicated focused 10-edge PAG is nevertheless threshold-sensitive, and some endpoints change with order or contradict chronology. Robust FCI$^+$ removes those literal reversals by retaining circles, at 1.64–1.85 times the paper-profile CI cost. Reliability improves; identifiability does not.

The hardest engineering work was translating ordered Possible-D-SEP pools and Algorithm 2 hierarchy/revisit semantics into executable invariants, then diagnosing finite-sample endpoint instability. Oracle-first validation, separating algorithm from application, and committing every seed and artifact worked well. Literal paper profiles did not provide sufficiently stable real-data endpoints, so that failure is reported rather than silently overridden with chronology. No method wins uniformly: paper FCI$^+$ is the fidelity and query-efficiency subject, while robust FCI$^+$ is preferred for this application under the declared audits.

The contribution is an auditable map of conditional-independence constraints and modelling sensitivity, not a rediscovered STAR causal model. Next steps are to reconstruct randomization strata for a formal intention-to-treat analysis; add cluster-aware CI tests and endpoint-level stability summaries; separate reading and mathematics; test temporal background knowledge as a distinct analysis; and derive target PAGs with an exhaustive MAG-equivalence oracle.

Reproduction. The `run_case_study` module rebuilds the STAR application with three benchmark repetitions, 100 discovery bootstraps, and 1,000 descriptive bootstraps; its refresh option re-executes the R reference. Scripts under `reports/` regenerate every figure, the frozen Figure 4(b) evidence, and the scaling audit. The committed `STAR_Students.tab.gz` has SHA-256 `2ffa578822eb30fcd6626fa7c5cc734a721fd05dd522403f0d643f89e891bac8`.[1]

References

Achilles, C. M., Bain, H. P., Bellott, F., Boyd-Zaharias, J., Finn, J., Folger, J., Johnston, J., and Word, E. (2008). Tennessee's Student Teacher Achievement Ratio (STAR) project. Harvard Dataverse, V1. `https://doi.org/10.7910/DVN/SIWH9F`.

Claassen, T., Mooij, J. M., and Heskes, T. (2013). Learning sparse causal models is not NP-hard. In *Proceedings of the 29th Conference on Uncertainty in Artificial Intelligence*.

causal-learn developers (2026). *FCI: Fast Causal Inference*. Official documentation, accessed 26 July 2026.

Glymour, C., Zhang, K., and Spirtes, P. (2019). Review of causal discovery methods based on graphical models. *Frontiers in Genetics*, 10, 524.

Kalisch, M., Mächler, M., Colombo, D., Maathuis, M. H., and Bühlmann, P. (2012). Causal inference using graphical models with the R package pcalg. *Journal of Statistical Software*, 47(11), 1–26.

Krueger, A. B. (1999). Experimental estimates of education production functions. *The Quarterly Journal of Economics*, 114(2), 497–532.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*, second edition. Cambridge University Press.

Richardson, T. S. and Spirtes, P. (2002). Ancestral graph Markov models. *The Annals of Statistics*, 30(4), 962–1030.

Spirtes, P., Glymour, C., and Scheines, R. (2000). *Causation, Prediction, and Search*, second edition. MIT Press.

Tetrad Project (2026). *Tetrad 7.6.10 release and FCI source*. Official CMU repository, accessed 26 July 2026.

Word, E., Johnston, J., Bain, H. P., Fulton, B. D., Zaharias, J. B., Achilles, C. M., Lintz, M. N., Folger, J., and Breda, C. (1990). *Student/Teacher Achievement Ratio (STAR): Tennessee's K–3 Class Size Study. Final Summary Report 1985–1990*. Tennessee State Department of Education.

Zhang, J. (2008). On the completeness of orientation rules for causal discovery in the presence of latent confounders and selection bias. *Artificial Intelligence*, 172(16–17), 1873–1896.

[1] This document is the standalone final report; every number is regenerated from committed artifacts by the listed scripts. Extended per-stage evidence tables and complete source locators are kept in the repository's full report, reports/fci_plus_star_report.tex.